

Trainings ▾

General Information

Exhaust gas is recirculated through the engine to reduce the amount of Oxides of Nitrogen (NO_x) emitted from the engine. The exhaust gas is cooled as it flows through the exhaust gas recirculation (EGR) cooler, and then is mixed with the compressed fresh air from the turbocharger before entering the intake manifold. EGR was introduced to reduce the amount of in-cylinder oxygen available for combustion while maintaining the same amount of flow through the engine. Exhaust gases present during the start of combustion are very stable and have a very slow reaction rate. They absorb heat during combustion, resulting in lower in-cylinder peak flame temperatures and therefore lower NO_x emissions.

The EGR mixer was designed to completely mix the EGR gas with the turbocharged air from the charge air cooler. Complete mixing is necessary to provide smooth operation and decreased emissions.

The EGR valve is controlled by the electronic control module (ECM). ISM with CM870 engines control the EGR valve by a pulse width modulated signal sent by the ECM and operates between 0 percent and 100 percent open. When the EGR valve is open, exhaust gas flows from the hot side of the engine, through the EGR cooler, and into the EGR mixer.

ISM with CM876 engines control the EGR valve through a J1939 datalink signal sent by the ECM and operates between 0 percent and 100 percent open. When the EGR valve is open, exhaust gas flows from the EGR cooler through the EGR connection tube, and into the EGR mixer.

The ISM with CM876 EGR connection tube allows the flow of cooled exhaust gas from the EGR valve outlet at the rear of the engine to the EGR mixer before the intake manifold. This engine utilizes a two-piece EGR connection tube design which also includes the EGR venturi tube.

The ISM CM870 and ISM CM875 EGR connection tube allows the flow of cooled exhaust gas from the EGR cooler outlet at the rear of the engine to the EGR mixer before the intake manifold. This engine utilizes a two-piece EGR connection tube which also includes the EGR venturi tube.

The EGR venturi tube contains a flow orifice that is used to measure the amount of exhaust gas flow using an EGR differential pressure sensor and a high and reference differential pressure sensing tubes.

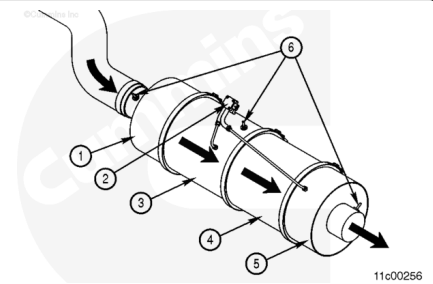
The exhaust manifold for the ISM with CM876 engine was redesigned to include a metal seal within the slip joints in efforts to reduce leakage. The exhaust manifold also has a port near the number one exhaust port for the exhaust pressure sensor.

The variable geometry turbocharger is required for several important engine operating conditions. It is used to facilitate EGR flow through the engine. Refer to Procedure 010-999 in Section F (</qs3/pubsys2/xml/en/procedures/35/35-010-999-tr.html>) for further information regarding variable geometry turbocharger and EGR flow interactions.

The variable geometry turbocharger is also used for engine braking. The variable geometry mechanism closes in order to increase exhaust manifold pressure. The increased exhaust pressure works against the pistons on the exhaust stroke retarding engine speed.

The aftertreatment system is used to reduce particulate emissions and is composed of six main components:

1. Aftertreatment inlet
2. Aftertreatment diesel particulate filter differential pressure sensor
3. Aftertreatment diesel oxidation catalyst
4. Aftertreatment diesel particulate filter
5. Aftertreatment outlet
6. Aftertreatment exhaust gas temperature sensors.

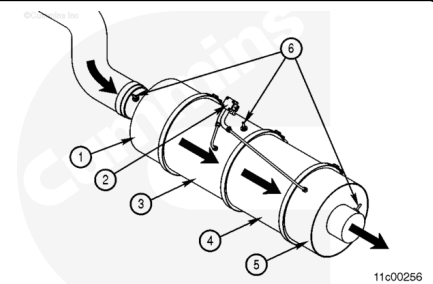


Note : In some applications, the aftertreatment diesel oxidation catalyst can be contained within the inlet section of the aftertreatment system.

Passive regeneration occurs when the exhaust temperatures are naturally high enough to oxidize the soot collected in the aftertreatment diesel particulate filter faster than the soot is collected.

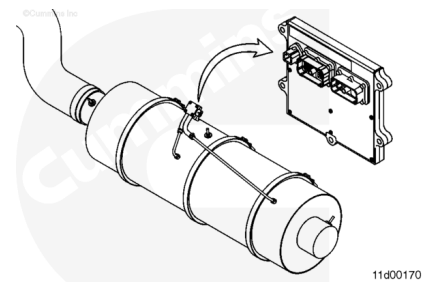
Passive regeneration typically occurs when the temperature of the aftertreatment diesel particulate filter is above 316°C [600°F]. This occurs during highway driving or driving with heavy loads.

Since passive regeneration occurs naturally, it is considered to be normal engine operation. No fuel is added to the exhaust stream during passive regeneration.



Active regeneration occurs when the exhaust temperatures are **not** naturally high enough to oxidize the soot collected in the aftertreatment diesel particulate filter faster than it is collected.

Active regeneration requires assistance from the engine in order to increase the exhaust temperature. This is typically done by injecting a small amount of diesel fuel into the exhaust stream (called aftertreatment injection) which is then oxidized by the aftertreatment diesel oxidation catalyst. The oxidation of this additional fuel creates the heat needed to regenerate the aftertreatment diesel particulate filter.



For active regeneration to occur, the ECM **must** detect that the aftertreatment diesel particulate filter restriction has reached a specified limit. Once this limit is reached, the engine will alter its operation in order to create exhaust temperatures high enough to actively regenerate the aftertreatment diesel particulate filter.

An active regeneration event typically consists of two parts: a warming up phase and a regenerating phase.

The purpose of the warm up phase is to increase the exhaust temperatures to the point that aftertreatment injection can occur.

Aftertreatment injection requires that temperatures in the aftertreatment system reach approximately 288°C [550°F]. At this temperature and above, the small quantities of fuel injected into the exhaust will properly oxidize across the aftertreatment diesel oxidation catalyst creating the additional heat required to actively regenerate the aftertreatment diesel particulate filter.

Once the warm up phase is complete and the aftertreatment injection has begun, the active regeneration phase begins.

During active regeneration, the engine ECM monitors the exhaust temperatures before and after the aftertreatment diesel particulate filter and maintains the temperatures in a range of approximately 427 to 649°C [800 to 1200°F]. The quantity of fuel used for aftertreatment injection will vary as the temperature is controlled within these limits.

The temperatures achieved during active regeneration are typically higher than those achieved during passive regeneration. The conversion of soot to carbon dioxide occurs much faster as temperatures increase.

A typical active regeneration event will take approximately 20 to 40 minutes to complete while the vehicle is operating. The vehicle operator may notice additional turbocharger noise during this time, along with an illuminated high exhaust temperature lamp, if equipped.

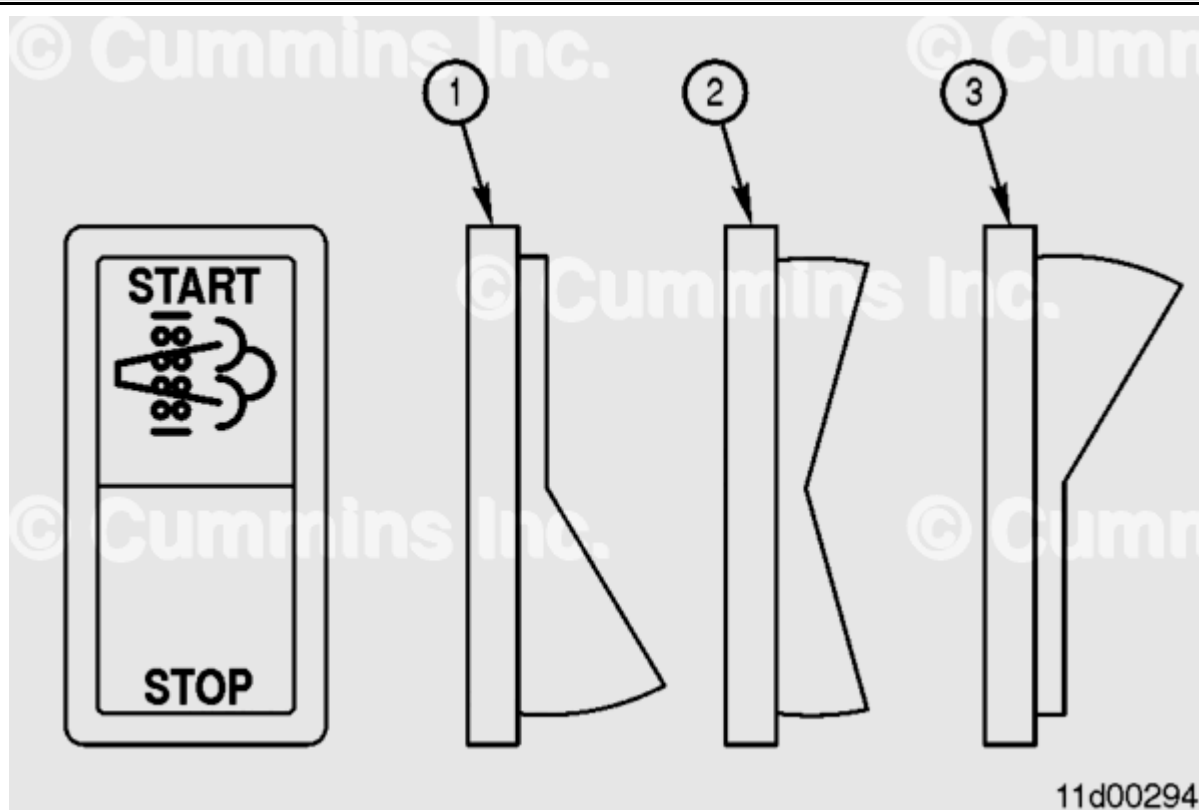
The frequency at which an engine will require an active regeneration varies greatly from application to application. In general, vehicles with a low vehicle speed, such as urban vehicles, or a low-load duty cycle, will require more active regeneration events than a heavily loaded vehicle or a vehicle with a high speed duty cycle.

The engine ECM also contains a time-based feature for active regenerations which is used to verify correct aftertreatment operation when the vehicle duty cycle is typically high enough that active regeneration events are **not** necessary.

Under some operating conditions, such as low speed, low load, or stop and go duty cycles, the engine may **not** have enough opportunity to regenerate the aftertreatment diesel particulate filter during normal vehicle operation. When this occurs, the engine illuminates the aftertreatment diesel particulate filter lamp to inform the vehicle operator that assistance is required, typically in the form of a stationary (parked) regeneration.

Stationary (parked) regeneration is a form of active regeneration that is initiated by the vehicle operator when the vehicle is **not** moving.

The following procedure provides more information on Stationary (Parked) regeneration. Refer to Procedure 014-013 in Section 14. (</qs3/pubsys2/xml/en/procedures/101/101-014-013.html>)



The vehicle manufacturer has the option of installing two switches that control aftertreatment functions: the start switch and the permit switch.

The start switch (known as the Diesel Particulate Filter Regeneration Start Switch in INSITE™ electronic service tool) is used to start a stationary (parked) regeneration. The vehicle manufacturer may also reference this switch as a stationary regeneration switch, start switch, or parked regeneration switch.

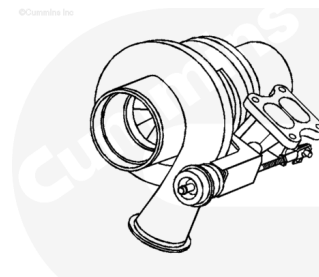
The permit switch (known as the Diesel Particulate Filter Permit Switch in INSITE™ electronic service tool) is used to allow the vehicle operator to disable active regeneration, if necessary. The vehicle manufacturer may also reference this switch as an inhibit switch, stop switch, or disable

switch.

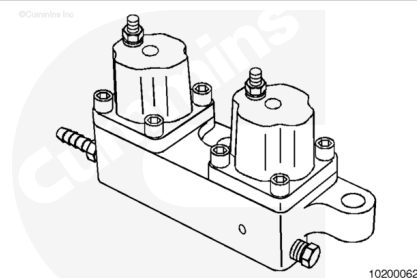
The following procedure describes the switch operation and troubleshooting steps. Refer to Procedure 011-056 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-056.html>)

Automotive with CM570

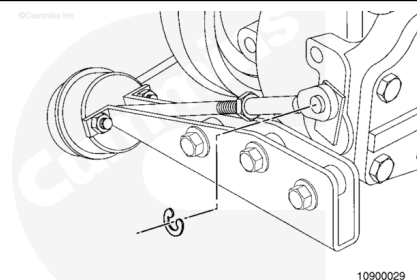
The wastegated turbocharger is a Holset® model HX55. It is comprised of a turbocharger, wastegate actuator, and wastegate valve in the turbine housing. A wastegated turbocharger provides improved response at low engine speeds without sacrificing turbocharger durability at high speeds. This is accomplished by allowing exhaust gases to bypass the turbine wheel during certain modes of engine operation. During low rpm operation, the turbocharger operates as a closed-system turbocharger, where the gases' energy is transferred to the compressor wheel and used to compress intake air. During high rpm operation, however, the turbocharger becomes an open-system turbocharger and allows exhaust gas to bypass the turbine. Since exhaust gas is gated around the turbine wheel, less energy is absorbed through the turbine and transferred to the compressor, reducing intake manifold pressures and turbine speeds.



The wastegate controller is mounted on the rear of the intake manifold on the turbocharger side of the engine and is controlled by the ECM. The wastegate controller regulates the percentage of the intake manifold pressure sent to the wastegate actuator. Two solenoids, similar to the fuel pump shutoff valves of the CELECT™ product, are used in conjunction with four orifices to regulate this percentage.



The wastegate actuator is mounted on the turbocharger and consists of a pressure canister, diaphragm, and rod. As pressure changes in the canister, as dictated by the wastegate controller, the actuator rod adjusts the wastegate valve accordingly.



The wastegate valve is mounted inside the turbocharger in the turbine housing. As the valve opens, exhaust gas is allowed to bypass the turbine wheel, lowering turbine speed to adjust intake manifold pressure.



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